

Forest of Undead

Game Design Document (Portfolio Edition)

Development Period: 4 January 2024 – 25 February 2024

Role: Solo Developer

Engine: Unity

Platform: PC / Web

Genre: 2D Top-Down Survival Shooter

Links:

Itch.io: <https://old-monkey.itch.io/forest-of-undead>

YouTube : <https://youtu.be/itYoGsx9BU?si=qjD1biM4c58GKEWz>

1. Game Overview

Forest of Undead is a fast-paced 2D top-down survival shooter inspired by endless survival games such as *Vampire Survivors*. The player is placed inside a cursed forest and must survive increasingly intense waves of undead enemies by continuously moving, fighting, and upgrading their abilities.

The game emphasizes long-term survival rather than completion. Failure is inevitable; the goal is to survive longer each run by making better upgrade decisions and adapting to escalating enemy pressure.



2. Core Fantasy & Design Pillars

Core Fantasy

Overwhelming power growth against endless undead hordes.

The player begins weak and vulnerable but gradually becomes stronger through upgrades, turning chaos into controlled destruction—until the game eventually overwhelms them.

Design Pillars

1. **Endless Pressure**

Enemy count, speed, and damage continuously increase, ensuring tension never plateaus.

2. **Build Expression**

Players shape each run through randomized upgrade choices, creating unique builds every playthrough.

3. **Readable Chaos**

Despite large enemy counts, visual clarity and predictable behavior ensure players understand why they fail.

4. **Fast Iteration**

Simple controls, short feedback loops, and immediate responses encourage repeated runs.

3. Core Gameplay Loop

The game follows a tightly focused survival loop:

Spawn Enemies → Defeat Enemies → Gain EXP → Level Up → Choose 1 of 3 Upgrades → Difficulty Scales → Repeat Until Death

As the loop progresses:

- Enemy spawn rates increase
- Enemy types expand
- Player decisions compound into stronger or weaker builds

This loop ensures replayability while reinforcing learning through failure.

4. Player Mechanics

Movement

- WASD movement
- Constant motion is required to avoid being surrounded

Combat

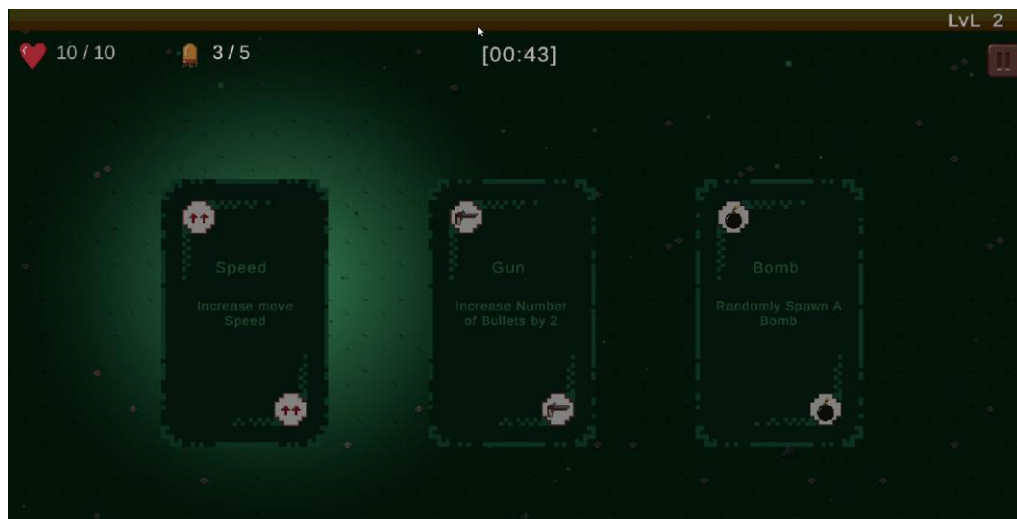
- Mouse input for shooting and reloading
- Toggleable auto-aim for accessibility and chaos management
- Area-control weapons supplement direct fire

Auto-aim exists to reduce mechanical fatigue and allow players to focus on positioning and upgrade strategy.

5. Progression & Upgrade System

Upon leveling up, the player chooses **1 of 3 randomized upgrades**. This system prioritizes replayability, long-term scalability, and ease of balancing. Upgrades are intentionally stat-driven to allow rapid iteration and clear cause–effect relationships for future balancing.

The upgrade system is documented in full for long-term maintainability and future reference.



Upgrade Categories & Detailed Effects

Bomb Upgrades (Area Burst Control)

- **Spawn Bomb:** Randomly spawns a bomb that explodes after a short delay

- **Bomb Count Increase:** Increases number of bombs spawned
- **Bomb Damage:** Increases bomb blast damage by **+10** per upgrade

Design Purpose: Clears dense enemy clusters and provides breathing room during late-game swarms.

Gun Upgrades (Primary DPS)

- **Bullet Count:** Increases bullets fired by **+2**
- **Gun Damage:** Increases shotgun damage by **+5**

Design Purpose: Supports high-risk, high-reward direct combat builds.

Gun Speed Upgrades (Fire Rate Optimization)

- **Reload Speed:** Decreases shotgun reload time by **-0.2f**
- **Bullet Speed:** Increases bullet travel speed by **+2**

Design Purpose: Improves responsiveness and sustained DPS output.

Heart Upgrades (Survivability)

- **Heal to Max:** Instantly restores current health to maximum
- **Max Health Increase:** Increases maximum health by **+1 heart**

Design Purpose: Allows recovery from mistakes and extends run longevity.

Scythe Upgrades (Orbiting Area Damage)

- **Add Scythe:** Unlocks the scythe weapon
- **Scythe Count:** Increases number of scythes
- **Scythe Damage:** Increases scythe damage by **+5**

Design Purpose: Provides passive close-range crowd control.

Speed Upgrades (Mobility)

- **Movement Speed:** Increases movement speed by **+0.2f**

Design Purpose: Enables escape from high-speed enemies and improves positioning.

Summon Upgrades (Indirect Damage)

- **Add Summon:** Unlocks a summon entity
- **Fireball Count:** Increases number of fireballs fired by summons
- **Fireball Damage:** Increases summon fireball damage by **+10**

Design Purpose: Provides consistent background damage with reduced player input.

Tornado Upgrades (Moving Area Denial)

- **Add Tornado:** Unlocks the tornado weapon
- **Tornado Count:** Increases number of active tornadoes
- **Tornado Damage:** Increases tornado damage by **+10**

Design Purpose: Controls space and forces enemies into predictable paths.

Upgrade Synergy & Build Expression

Although upgrades are stat-based, their combinations naturally form build archetypes:

- **Area Control Build:** Tornado + Scythe + Bombs
- **Summoner Build:** Summons + Speed + Health
- **Glass Cannon Build:** Gun Damage + Gun Speed + Movement Speed

This system rewards commitment to a playstyle while preserving randomness for replayability.

Each run feels distinct based on upgrade combinations and player decision-making.

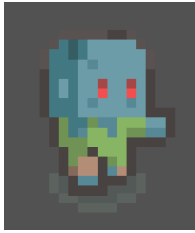
6. Enemy Design

Enemy Design Philosophy

Enemy difficulty scales through **controlled stat progression** instead of complex AI behaviors. This ensures clarity, performance stability, and predictable escalation in an endless survival format.

Enemy Roles

Enemy 0 – Starter Undead



Role: Baseline threat

Introduces core combat mechanics and movement without overwhelming pressure.

Enemy 1 – Fast Undead



Role: Reaction pressure

Forces players to stay mobile and discourages static positioning.

Enemy 2 – Durable Runner



Role: Mid-game pressure unit

Tests damage scaling and exposes weak builds.

Enemy 3 – Aggressive Runner



Role: Attrition amplifier

Overwhelms players through speed, accelerating failure if positioning or DPS is insufficient.

Enemy 4 – Heavy Hitter



Role: Positional punishment unit

Slow but extremely punishing on contact, rewarding careful movement.

Enemy Stats

ENEMY	HEALTH	DAMAGE	SPEED
ENEMY 0	10	1 Heart	1.0
ENEMY 1	10	1 Heart	1.5
ENEMY 2	15	2 Hearts	2.0
ENEMY 3	15	2 Hearts	3.0
ENEMY 4	20	3 Hearts	1.0

7. Difficulty Scaling

Difficulty increases over time through:

- Faster enemy spawn intervals
- Expanded enemy pools
- Increased screen saturation

This creates a natural progression from learning → mastery → inevitable failure.

8. UI, Audio & Visual Design

UI

- Health displayed as hearts for clarity
- EXP bar communicates progression pacing
- Level timer reinforces survival pressure



Visual Style

- Pixel art prioritizing contrast and readability
- Minimal visual noise during high-intensity moments

Audio

- Sound cues provide feedback for danger, damage, and upgrades
 - Background music maintains tension without overwhelming gameplay
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9. Production Timeline

- **4 Jan – Mid Jan:** Core movement, shooting, enemy spawning
- **Mid Jan – Early Feb:** Upgrade system, weapons, enemy scaling
- **Early Feb – 25 Feb:** UI, polish, audio, balancing, playtesting

The scope was intentionally limited to ensure full completion as a solo developer.

10. Postmortem

What Went Right

- Complete gameplay loop achieved within time constraint
- High replayability through upgrade randomness

What Could Be Improved

- Enemy behavior variety is limited to stat scaling
- Additional enemy modifiers could add depth without heavy AI complexity

Future Improvements

- Introduce behavioral modifiers (slow, knockback resistance)
 - Add meta-progression between runs
 - Expand visual variety while maintaining readability
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Credits

- Character And environment Sprite: Sourced from free game asset packs (creator unknown — will credit properly if identified)
- Summon , Tornado And Bomb Animation : Dracula
- Cards And Ui Design : Kash
- Sound Effects : <https://freesound.org> and <https://pixabay.com>

- Background music : <https://freesound.org> and <https://pixabay.com>
- Ui images : freepik
- Game Design & Programming: Kash

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Forest of Undead demonstrates end-to-end solo development, from concept to playable experience, with a strong focus on systems design, scalability, and player readability.